

ACKNOWLEDGEMENT

It has been rightly said that we are built on the shoulders of others. For everything I have achieved, the credit goes to all those who have helped me to complete this internship training successfully.

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ABSTRACT

This internship report presents the learning experience and practical knowledge gained during the internship at TechieHelp, an AI-based software development company. The internship was carried out from 27 January 2026 to 13 March 2026, where I worked as a Full Stack Developer Intern.

During the internship, I gained hands-on experience in modern web development technologies such as HTML, CSS, JavaScript, React.js, Node.js, Express.js, and MongoDB. The training focused on understanding the software development lifecycle, including designing, coding, testing, debugging, and deployment of web applications.

As part of the internship, I developed a project titled “TalentIQ – AI-Based Code Execution Platform.” This project allows users to write and execute code in multiple programming languages in real time through a web-based interface. It helped in understanding full-stack development and API integration.

The internship also enhanced my problem-solving abilities, debugging skills, and understanding of real-world development environments. Additionally, I developed professional skills such as teamwork, communication, and time management.

Overall, this internship played a significant role in improving my technical knowledge and preparing me for a career in software development.

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CHAPTER 1 – INTRODUCTION

1.1 Background of the Internship

Internships are an essential part of professional education, especially in the field of Computer Applications and Information Technology. They serve as a bridge between theoretical knowledge gained in academic institutions and practical exposure required in the industry. While classroom learning provides a strong foundation of concepts, it is through internships that students gain real-world experience and understand how these concepts are applied in practical scenarios.

In today's rapidly evolving technological landscape, the demand for skilled software professionals has increased significantly. Organizations seek individuals who are not only proficient in theoretical concepts but also capable of working on real-time projects, solving complex problems, and adapting to dynamic work environments. Internships provide an opportunity for students to develop these skills by working in professional settings.

The internship training allows students to gain hands-on experience in modern technologies, tools, and frameworks used in software development. It also helps in understanding the software development lifecycle, including requirement analysis, system design, implementation, testing, and deployment. Through this exposure, students become familiar with industry standards, coding practices, and collaborative workflows.

Moreover, internships contribute to the development of professional skills such as communication, teamwork, time management, and responsibility. Working under the guidance of experienced mentors and collaborating with team members enhances confidence and prepares students to face real-world challenges.

The internship undertaken at TechieHelp provided an excellent opportunity to explore the practical aspects of full-stack web development and gain exposure to modern technologies such as MERN stack. It helped in strengthening technical knowledge and developing a problem-solving mindset, which is essential for a successful career in the IT industry.

1.2 Company Profile



Figure 1: TechieHelp Company Logo

TechieHelp is an emerging technology-driven organization specializing in software development and artificial intelligence solutions. Established in the year 2023, the company focuses on delivering innovative and efficient digital products that address real-world problems using modern technologies.

The organization operates in multiple domains including full-stack web development, artificial intelligence, machine learning, and software engineering. TechieHelp emphasizes research and development to create scalable, user-friendly, and high-performance applications. The company's approach is centered around innovation, creativity, and continuous improvement.

The vision of TechieHelp is to become a globally recognized technology company that contributes to digital transformation through intelligent software solutions. Its mission includes developing high-quality applications, promoting innovation in artificial intelligence, and providing practical learning opportunities to students and aspiring developers.

TechieHelp follows a structured organizational model consisting of various teams such as management, development, research and development, and interns. Each team works collaboratively to achieve project objectives and deliver quality solutions. The development team focuses on designing and implementing software systems, while the research team explores new technologies and innovative ideas.

The company provides a dynamic and learning-oriented work environment where interns are encouraged to participate in real-time projects, technical discussions, and collaborative tasks. This environment helps in enhancing both technical and professional skills.

One of the key highlights of TechieHelp is its internship program, which is designed to provide students with hands-on experience in modern software development practices. The program focuses on practical learning, problem-solving, and exposure to industry-standard tools and technologies. Interns are guided by experienced mentors who help them understand development workflows and improve their coding skills.

Overall, TechieHelp plays a significant role in shaping the skills of aspiring developers by providing them with opportunities to work on real-world projects and gain valuable industry experience.

1.3 Internship Overview

The internship at TechieHelp was carried out from 27 January 2026 to 13 March 2026 as a Full Stack Developer Intern. The primary aim of the internship was to gain practical exposure to modern web development technologies and understand how software applications are developed in real-world environments.

During the internship, various technologies such as HTML, CSS, JavaScript, React.js, Node.js, Express.js, and MongoDB were introduced and practiced. These technologies are widely used in the development of modern web applications and form the foundation of full-stack development.

The internship program was structured to include both theoretical learning and practical implementation. Initially, the focus was on understanding the fundamentals of web development, including the structure and design of web pages. Gradually, the training moved towards advanced concepts such as backend development, API integration, and database management.

As part of the internship, a project titled “TalentIQ – AI-Based Code Execution Platform” was developed. This project involved creating a web-based platform where users can write and execute code in multiple programming languages. The project provided hands-on experience in integrating frontend and backend technologies and handling real-time data processing.

In addition to technical learning, the internship also involved participation in team discussions, problem-solving sessions, and debugging activities. These activities helped in improving analytical thinking and understanding real-world development challenges.

The internship provided valuable exposure to the complete software development lifecycle and helped in understanding how different components of a system work together to deliver a functional application. It also enhanced the ability to work in a team and adapt to a professional work environment.

1.4 Objectives of the Internship

The primary objective of the internship was to bridge the gap between academic learning and practical implementation by providing hands-on experience in software development.

The key objectives of the internship are as follows:

- To understand the complete software development lifecycle, including planning, designing, coding, testing, and deployment of applications.
- To gain practical knowledge of full-stack development using MERN stack technologies such as MongoDB, Express.js, React.js, and Node.js.
- To enhance programming skills and develop the ability to write efficient and structured code.
- To learn the use of development tools such as Visual Studio Code, Git, and GitHub for coding, debugging, and version control.
- To improve problem-solving and debugging skills through real-time development tasks.
- To understand the importance of teamwork, communication, and collaboration in a professional environment.
- To build confidence in handling real-world projects and preparing for future career opportunities in the IT industry.

The internship successfully achieved these objectives by providing practical exposure to modern technologies and helping in the overall development of technical and professional skills.

CHAPTER 2 – INTERNSHIP DETAILS

2.1 Internship Information

The internship was carried out at TechieHelp, an AI-based software development company, with the objective of gaining practical knowledge in full-stack web development. The internship provided exposure to modern technologies and real-world development practices.

Particulars	Details
Company Name	TechieHelp – AI Software Development Company
Internship Role	Full Stack Developer Intern
Duration	27 January 2026 – 13 March 2026
Department	Software Development
Internship Mode	Online / Remote
Mentor Name	Er. Aditya Kumar
Academic Guide	Prof. Gourambika D
Tools Used	VS Code, Git, GitHub
Technologies Used	MERN Stack (MongoDB, Express, React, Node)

Table 1: Iternship Informations

2.2 Roles and Responsibilities

During the internship at TechieHelp, I worked as a Full Stack Developer Intern and was actively involved in various development activities. The internship provided an opportunity to work on both frontend and backend technologies and understand the complete development process.

The key roles and responsibilities performed during the internship are as follows:

- Developing user interfaces using HTML, CSS, and React.js to create responsive and user-friendly web applications.
- Writing backend logic using Node.js and Express.js to handle server-side operations and API development.
- Working with MongoDB for storing and managing application data efficiently.
- Designing and implementing application features as part of project development.

- Debugging and fixing errors in code to ensure smooth functioning of applications.
- Using Git and GitHub for version control and managing project code.
- Collaborating with mentors and fellow interns to understand project requirements and improve development skills.
- Testing applications to ensure proper functionality and user experience.

These responsibilities helped in gaining practical experience in full-stack development and improved both technical and problem-solving skills.

2.3 Tools and Technologies Used

During the internship, various tools and technologies were used to develop web applications and understand real-world software development practices. These technologies form the foundation of modern web development.



Figure 2: Technology Stack Architecture

Frontend Technologies:

- HTML – Used for structuring web pages.
- CSS – Used for designing and styling user interfaces.
- JavaScript – Used for adding interactivity and dynamic behavior.
- React.js – Used for building modern and component-based user interfaces.

Backend Technologies:

- Node.js – Used for server-side programming.
- Express.js – Used for building APIs and handling backend logic.

Database:

- MongoDB – Used for storing and managing application data.

Development Tools:

- Visual Studio Code – Code editor used for development.
- Git – Version control system.
- GitHub – Platform for code hosting and collaboration.

API Communication:

- Axios – Used for sending HTTP requests between frontend and backend

These tools and technologies helped in understanding how modern web applications are developed and deployed efficiently.

2.4 Development Environment

The development environment plays a crucial role in software development as it provides the necessary tools and setup required for building applications.

During the internship, Visual Studio Code was used as the primary code editor due to its flexibility, extensions, and debugging features. Git and GitHub were used for version control and collaboration, allowing efficient management of code changes.

The frontend development was carried out using React.js, which provided a component-based architecture for building user interfaces. The backend was developed using Node.js and Express.js, which enabled the creation of APIs and server-side logic.

MongoDB was used as the database to store application data in a flexible and scalable format. The integration between frontend and backend was achieved using Axios for API communication.

This development environment helped in understanding the complete workflow of building, testing, and deploying modern web applications.

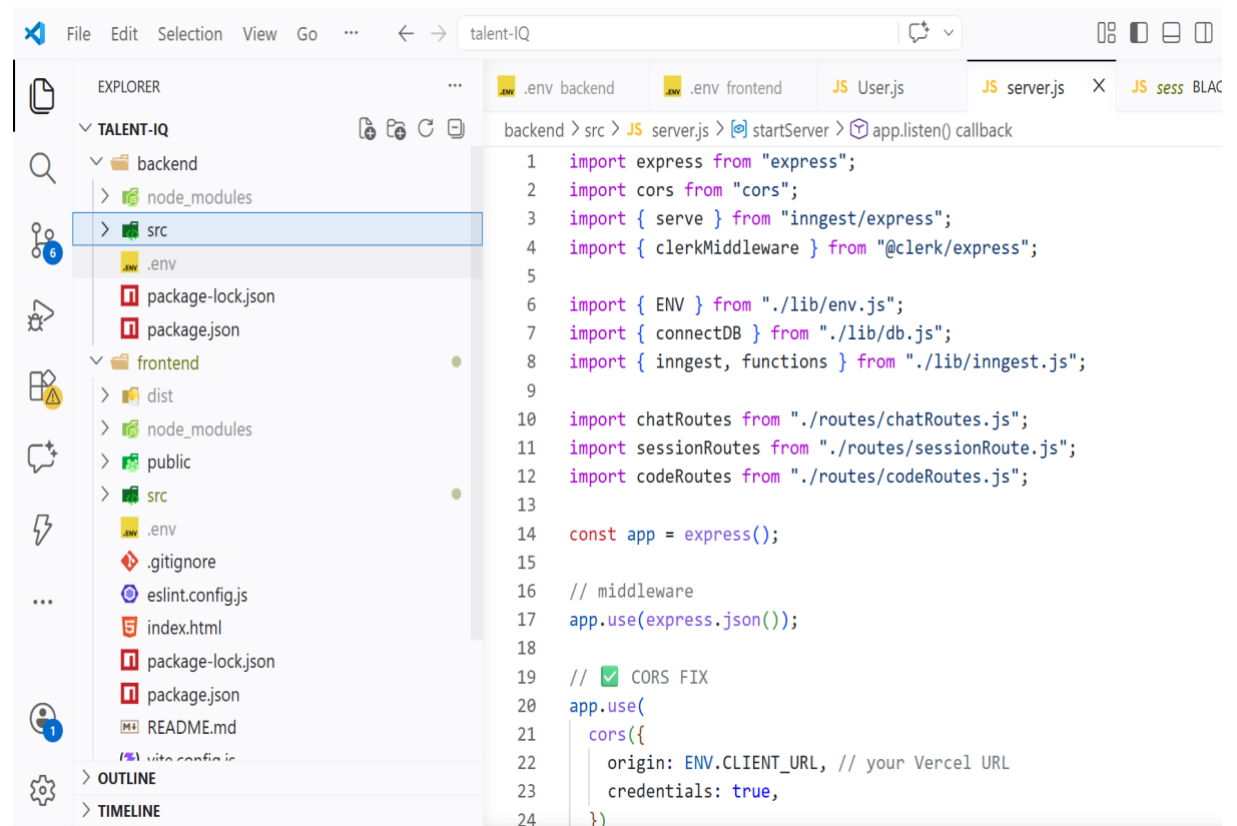


Figure 3: Development Environment

CHAPTER 3 – WORK / TASKS UNDERTAKEN

3.1 Introduction to Work

During the internship at TechieHelp, various practical development tasks were performed to gain hands-on experience in full-stack web application development. The work focused on understanding real-world system architecture, API integration, and building scalable applications using modern technologies.

As part of the internship, a major project titled “TalentIQ – AI-Based Code Execution Platform” was developed. The project is a full-stack web application that allows users to write, execute, and manage code in multiple programming languages in real time.

The development involved working on frontend, backend, and database layers along with integrating external APIs for code execution and real-time communication. The project follows a modular and scalable architecture, ensuring efficient performance and seamless user experience.

This work provided deep understanding of how modern web applications are structured, deployed, and managed in real-world environments.

3.2 Problem Statement

In the current digital learning environment, users require platforms that allow them to practice coding efficiently without complex setup processes. Traditional coding environments require installation of compilers, configuration of dependencies, and system-level setup, which can be challenging for beginners.

Additionally, there is a lack of integrated platforms that combine code execution, session management, and real-time collaboration features in a single system.

Therefore, the problem is to design and develop a web-based platform that enables users to write, execute, and manage code seamlessly while supporting real-time interaction and efficient backend processing.

3.3 Proposed Solution

To address the identified problem, a full-stack web application named “TalentIQ” was developed. The system provides a browser-based platform where users can write and execute code in real time without requiring local setup.

The platform is built using a multi-layer architecture consisting of frontend, backend, database, and external services. The frontend provides an interactive user interface, while the backend handles API requests, authentication, and execution logic.

External APIs such as JDoodle are integrated for code execution, and Stream API is used for real-time communication features. The system ensures secure and efficient handling of user requests and data.

This solution simplifies coding practice and provides a scalable and user-friendly environment for developers and learners.

3.4 System Architecture

The TalentIQ platform follows a layered architecture consisting of frontend, backend, database, and external services. Each layer is responsible for specific functionalities and communicates with other layers through APIs.

The frontend layer is built using React.js (Vite) and provides UI components such as code editor, output panel, dashboard, and session management interface. Clerk is used for authentication and user management.

The backend layer is developed using Node.js and Express.js, which handle REST APIs such as `/api/sessions`, `/api/code/run`, and `/api/chat`. The backend also includes middleware for authentication and business logic for session management and code execution.

The database layer uses MongoDB Atlas to store user data, sessions, and messages efficiently.

External services such as JDoodle API are used for code execution, while Stream API enables real-time chat and communication.

The deployment is handled using platforms such as Vercel for frontend, Railway for backend, and MongoDB Atlas for database hosting.

This architecture ensures scalability, modularity, and efficient performance of the system.

System Architecture of TalentIQ Platform:

The above figure illustrates the overall architecture of the TalentIQ platform, showing the interaction between frontend, backend, database, and external services. The frontend is built using React.js (Vite) for user interaction, while the backend uses Node.js and Express.js to handle API requests and business logic. MongoDB Atlas is used for data storage, and external services like JDoodle API and Stream API are integrated for code execution and real-time communication. The system is deployed using Vercel and Railway, ensuring scalability and efficient performance.

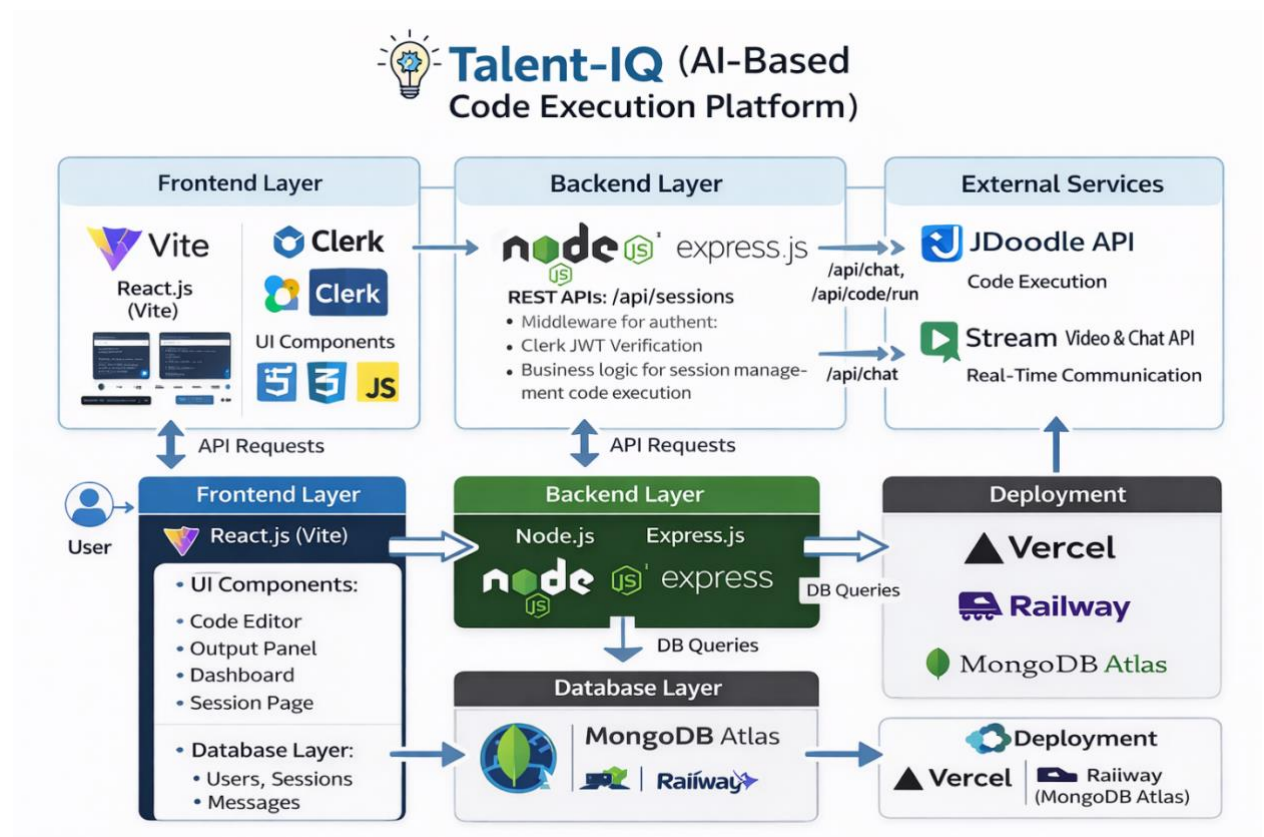


Figure 4: System Architecture of TalentIQ Platform

Workflow Diagram of Talent-IQ

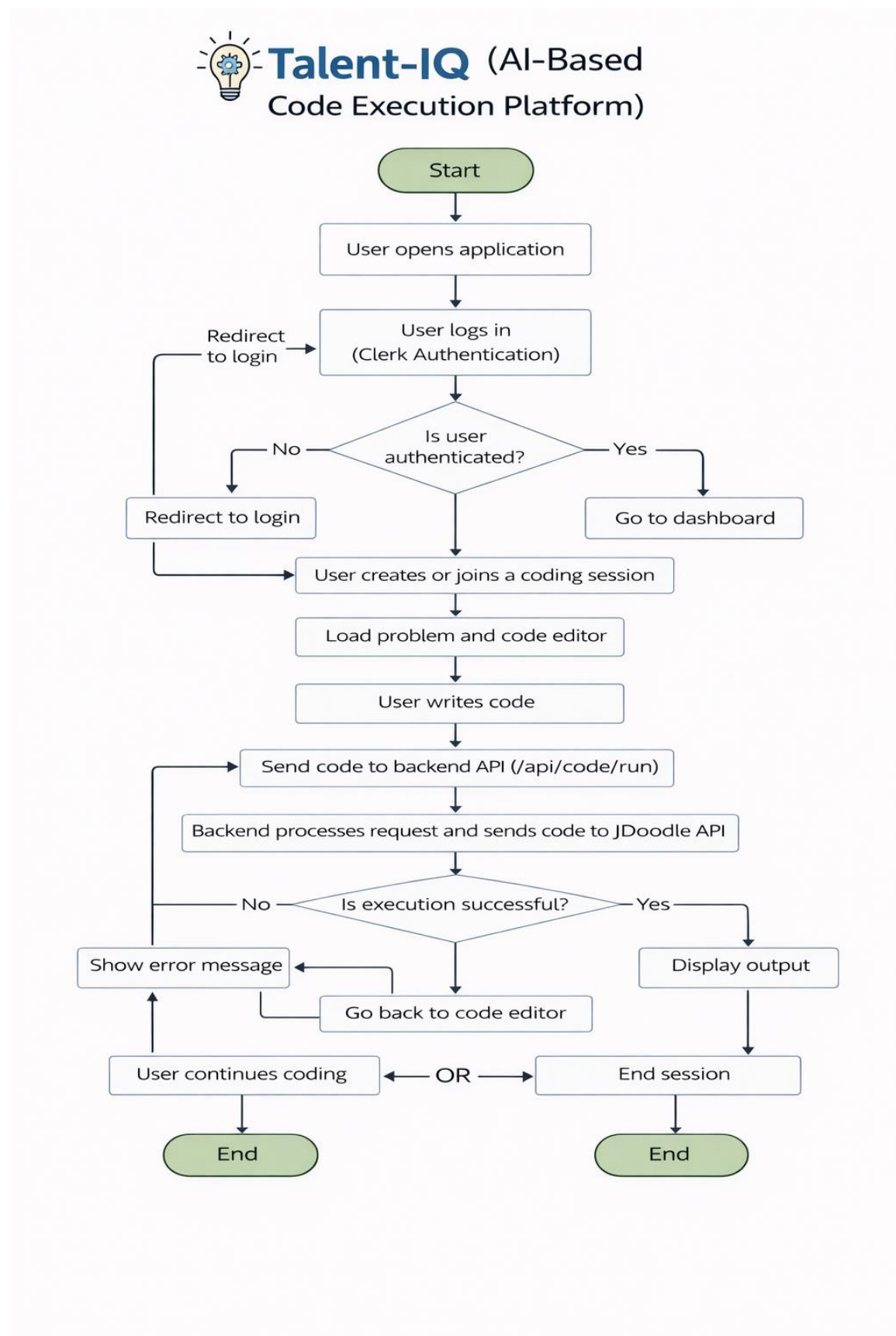


Figure 5: Flowchart of Talent-IQ

3.5 Features of the System

The TalentIQ platform includes several advanced features that enhance user experience and functionality. Each feature is designed to provide an efficient and interactive coding environment. The major features of the system are described below:

The above figure illustrates the overall design and functionality of the TalentIQ platform, which is developed using the MERN stack (MongoDB, Express.js, React.js, and Node.js). The platform is designed to simulate a real-time remote coding interview environment.

It integrates multiple features such as an interactive code editor, real-time code execution, and live communication through video and chat. Users can participate in coding sessions, solve problems, and collaborate with others in a seamless and efficient manner.

The interface highlights the modern UI design along with backend integration and external services, providing a complete full-stack solution. This system demonstrates how frontend, backend, and database technologies work together to build a scalable and user-friendly application.

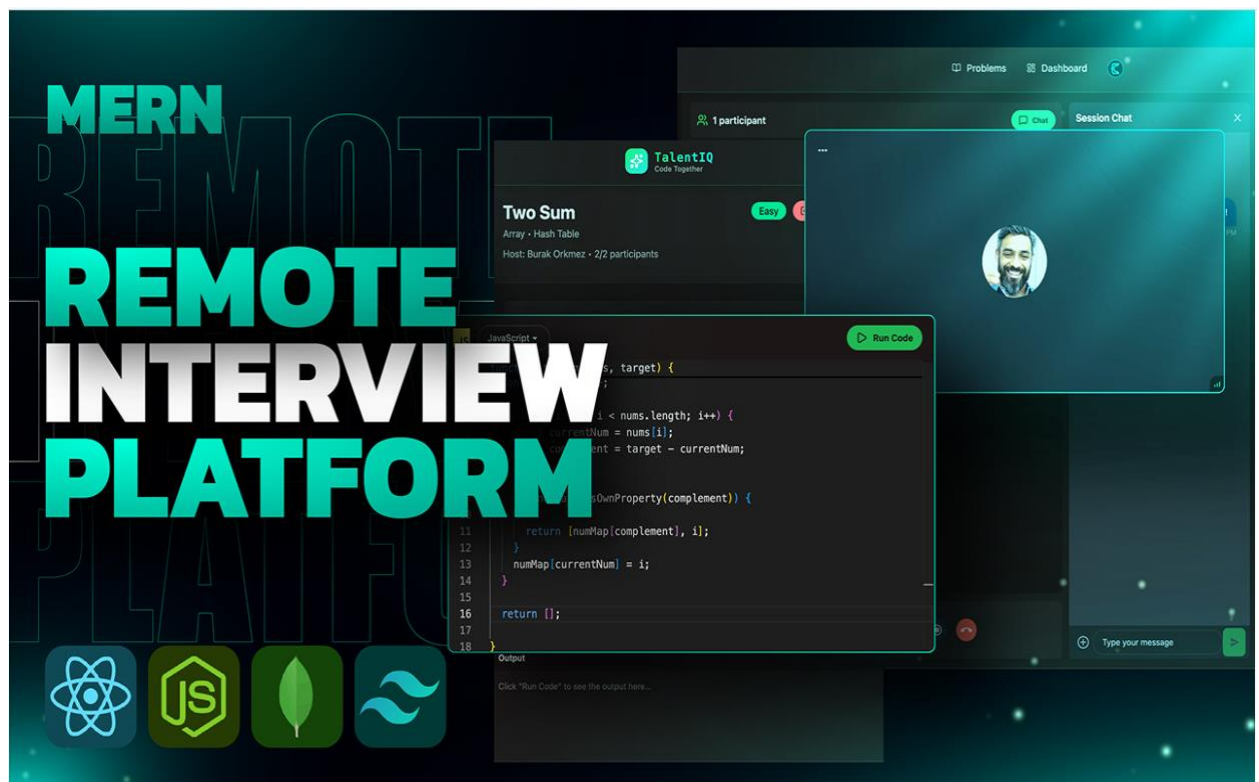


Figure 6: Hero Banner Of Talent-IQ

3.5.1 User Authentication (Clerk Integration)

The platform provides secure user authentication using Clerk. Users can sign up and log in to access the system. Authentication ensures that only authorized users can create or join coding sessions and access their data securely.

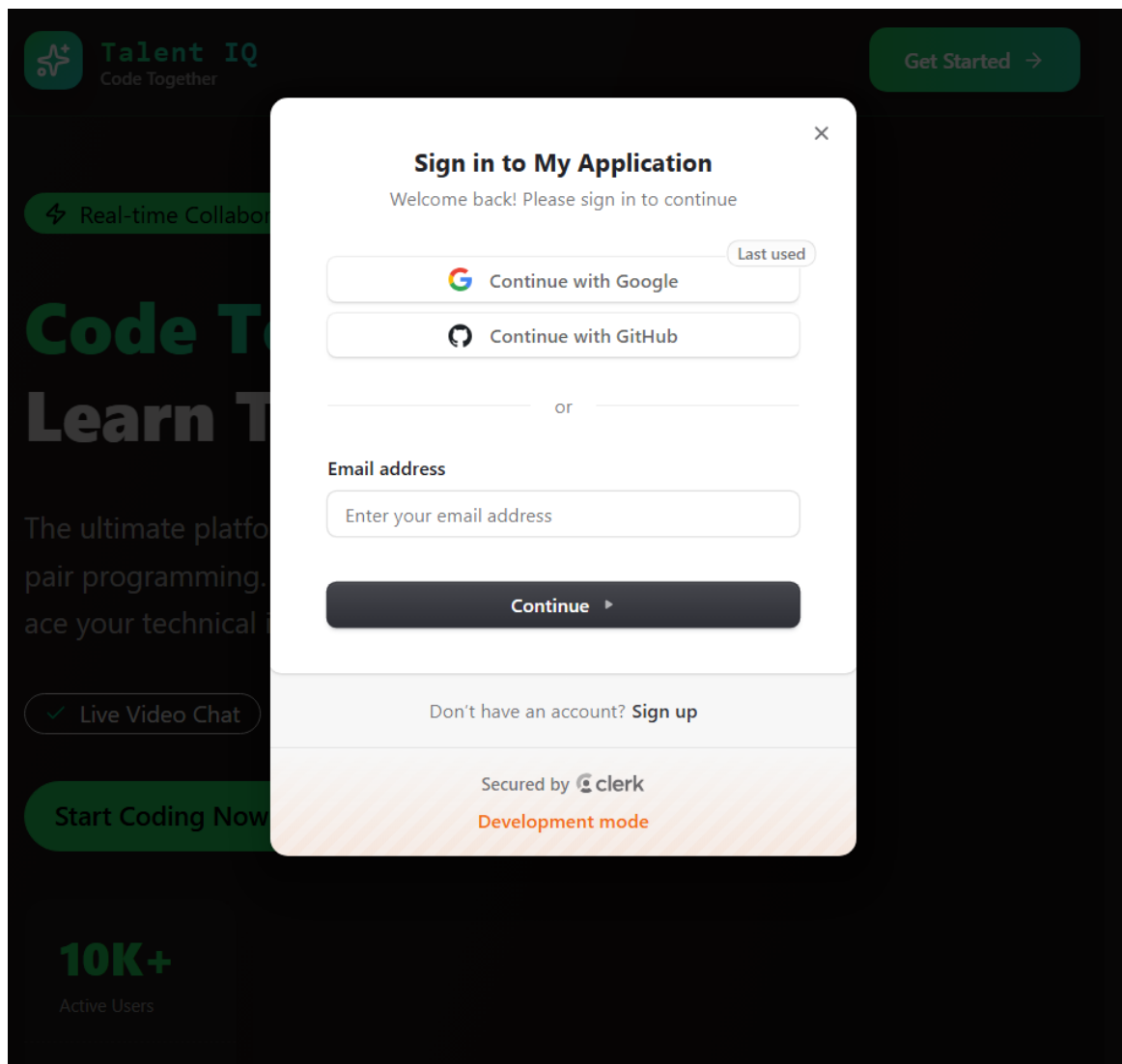


Figure 7: User Authentication (Clerk Integration)

3.5.2 Dashboard for Session Management

The dashboard provides an overview of active coding sessions. Users can create new sessions or join existing ones. It acts as the central control panel for managing coding activities.

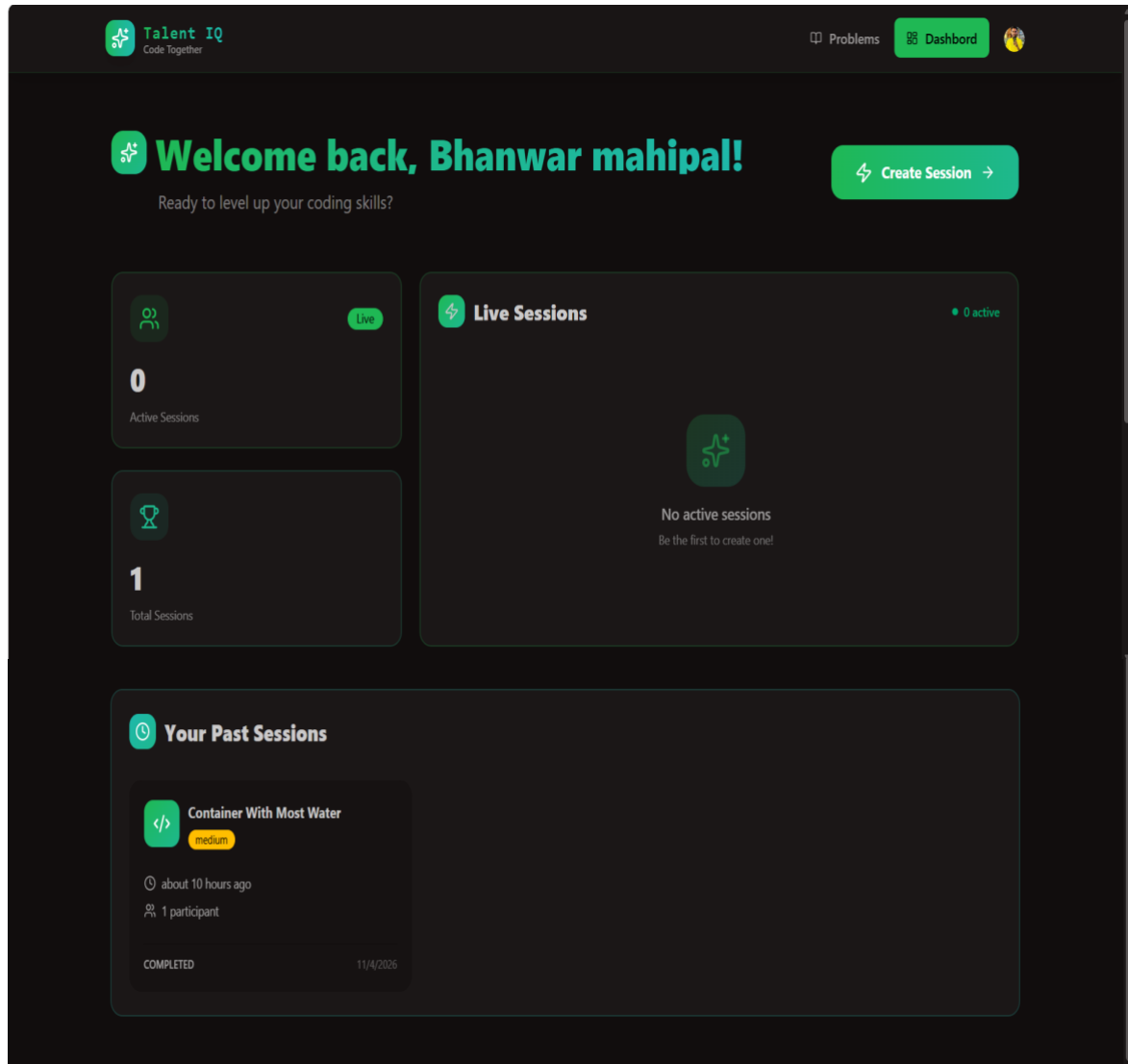


Figure 8: Dashboard for Session Management

3.5.3 Interactive Code Editor

The platform includes an interactive code editor where users can write, edit, and manage their code efficiently. It provides a user-friendly interface with support for multiple programming languages.

Support multiple languages like JavaScript, java and Python.

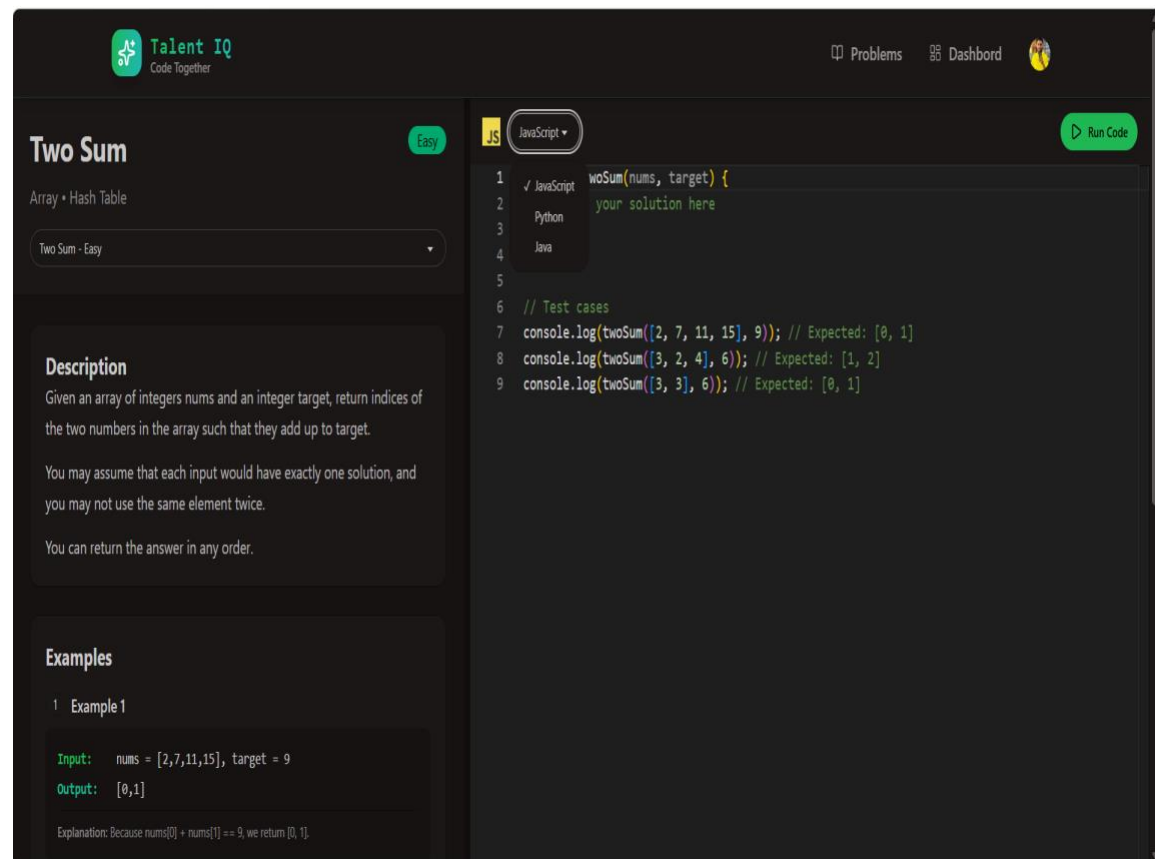


Figure 9: Interactive Code Editor

3.5.4 Practice Problems Page

The TalentIQ platform includes a dedicated Practice Problems page that allows users to enhance their coding skills by solving structured programming problems. This section provides a collection of curated problems categorized based on difficulty level and topic, such as arrays, strings, and data structures.

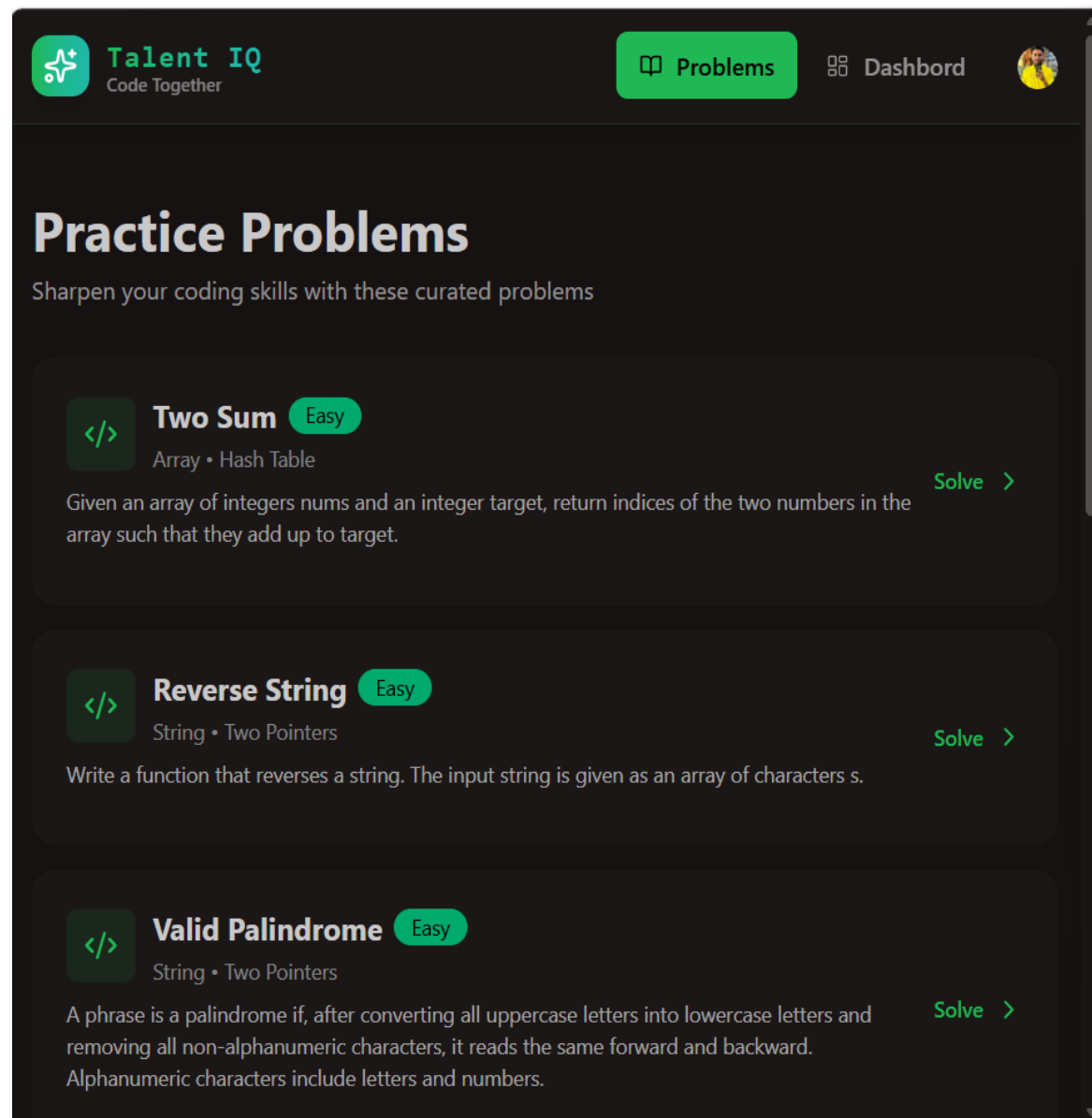


Figure 10: Practice Problems Page

3.5.5 Session-Based Coding System

Users can create or join coding sessions, allowing them to work collaboratively or individually. Sessions help in organizing coding activities and maintaining user progress.

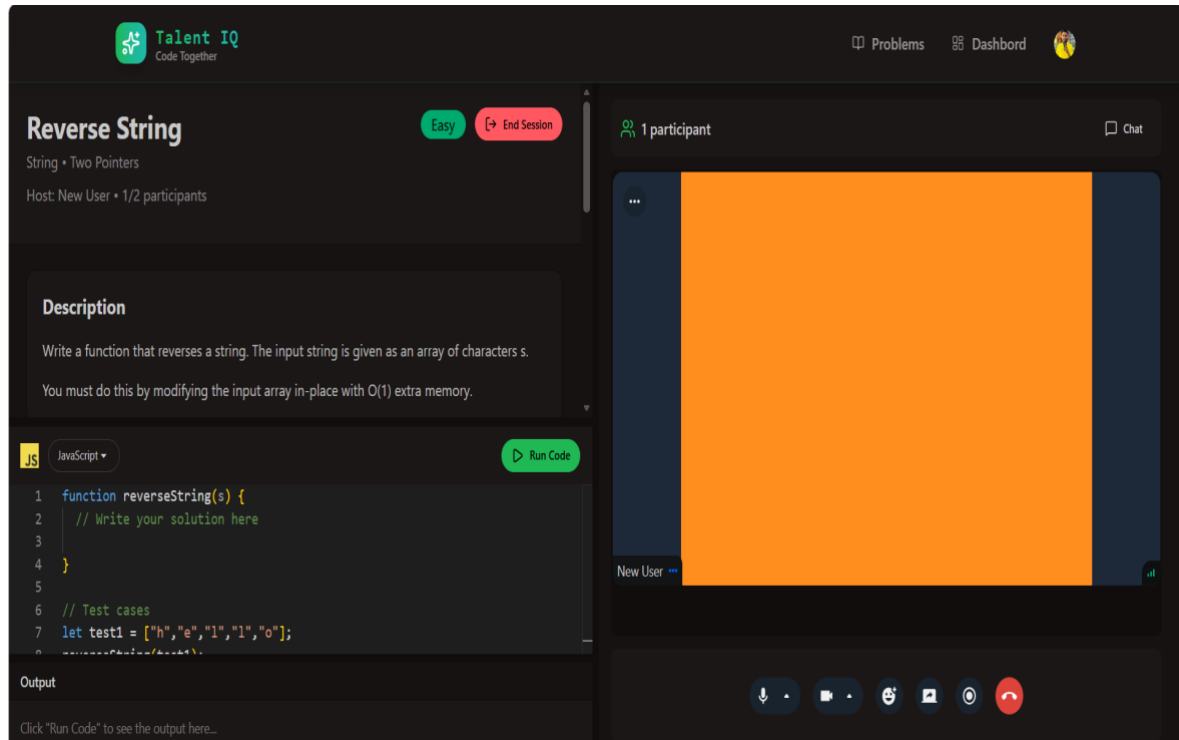


Figure 11: Session-Based Coding System

3.5.6 Real-Time Communication (Stream API)

The platform supports real-time communication using Stream API. Users can interact through chat or video call features, enabling collaborative coding and discussion.

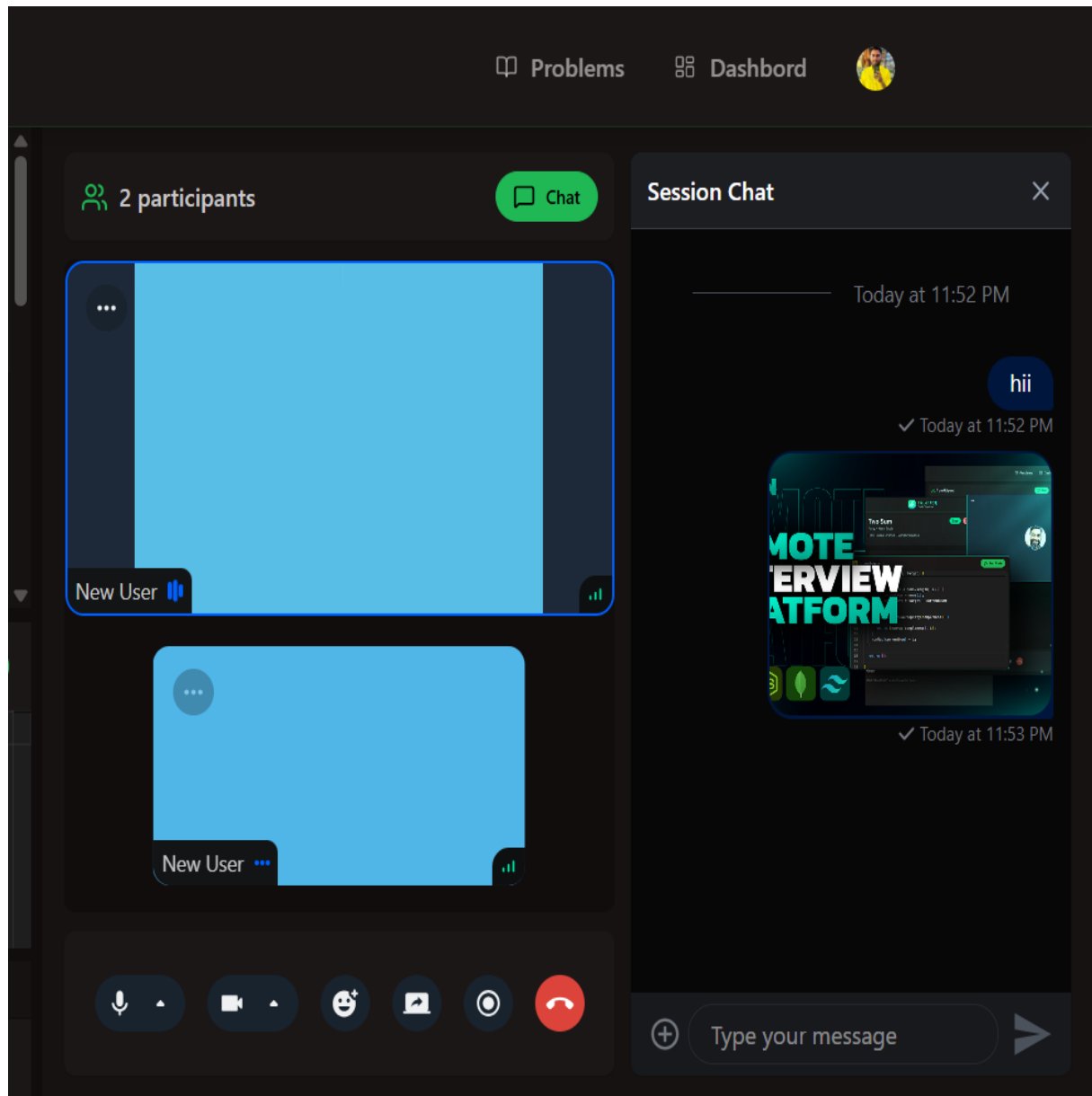


Figure 12: Real-Time Communication (Stream API)

3.6 Code Implementation

The implementation of the TalentIQ platform involves communication between frontend and backend using REST APIs. Axios is used to send HTTP requests from the frontend to the backend server.

The backend processes the request, interacts with external APIs for code execution, and returns the output to the frontend.

The following code snippet demonstrates API communication:

Piston.js

```
// 🚀 Backend proxy execution (FINAL FIXED)

const BACKEND_API = "http://localhost:3000/api/code";

const LANGUAGE_VERSIONS = {

  javascript: { language: "javascript", version: "18.15.0" },

  python: { language: "python", version: "3.10.0" },

  java: { language: "java", version: "15.0.2" },

};

// ✅ ADD problemId parameter

export async function executeCode(language, code, problemId) {

  try {

    const languageConfig = LANGUAGE_VERSIONS[language];

    if (!languageConfig) {

      return {

        success: false,

        error: `Unsupported language: ${language}`,

      };

    }

    const response = await fetch(`${BACKEND_API}/run`, {
```



```
method: "POST",

headers: {

  "Content-Type": "application/json",

},

body: JSON.stringify({

  language: languageConfig.language,

  code: code,

  problemId: problemId, // 🔥 IMPORTANT FIX

}),

});

if (!response.ok) {

  const text = await response.text();

  console.error("API ERROR:", text);

  return {

    success: false,

    error: `HTTP error! status: ${response.status}`,

  };

}

const data = await response.json();

console.log("BACKEND RESPONSE:", data);

return {

  success: data.passed,

  output: data.output || "No output",

  error: data.passed ? null : "Tests failed",

};
```

```

    } catch (error) {

        console.error("EXECUTION ERROR:", error);

        return {

            success: false,

            error: `Failed to execute code: ${error.message}`,

        };

    }

}

```

3.7 Database Design

The database design of the TalentIQ platform is implemented using MongoDB Atlas. The database stores user-related data, session information, and communication messages.

This structure ensures efficient data storage and retrieval while maintaining scalability.

Collection Name	Field Name	Data Type	Description
Users	_id	ObjectId	Unique identifier for user
Users	name	String	Name of the user
Users	email	String	User email address
Users	password	String	Encrypted user password
Users	createdAt	Date	Account creation timestamp
Sessions	_id	ObjectId	Unique session identifier

Table 2: Database Structure

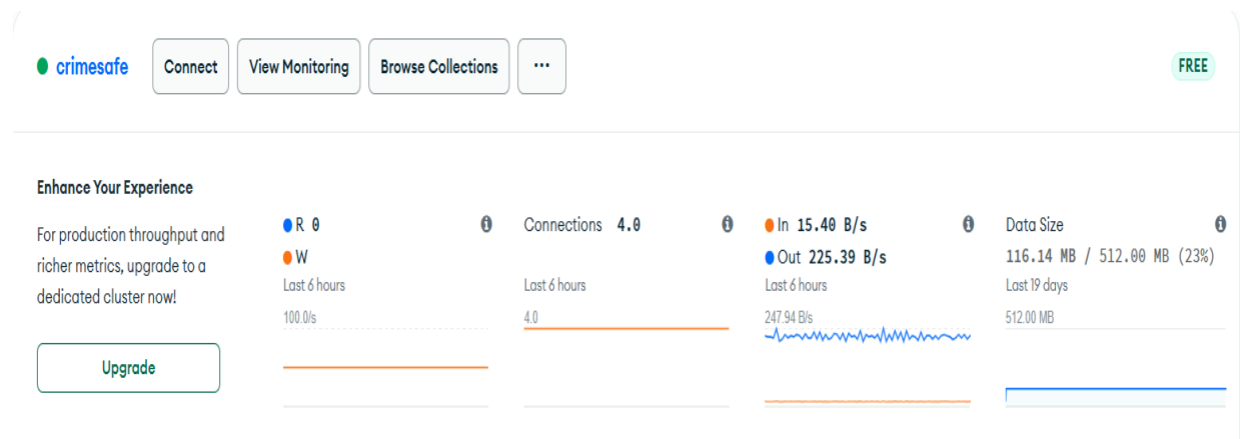


Figure 13: Backend API Execution

3.8 Tasks Performed During Internship

During the internship, several development tasks were performed:

- Developed frontend components using React.js
- Implemented backend APIs using Node.js and Express.js
- Integrated Clerk authentication system
- Connected MongoDB database for data storage
- Integrated JDoodle API for code execution
- Implemented real-time chat using Stream API
- Debugged and optimized application performance
- Deployed application using Vercel and Railway

These tasks helped in gaining real-world experience in building scalable full-stack applications.

CHAPTER 4 – LEARNING OUTCOMES

4.1 Introduction

The internship at TechieHelp provided valuable learning opportunities that contributed significantly to both technical and professional development. Through hands-on experience, real-world project work, and interaction with mentors, various important skills were acquired during the internship period.

The learning outcomes of this internship are not limited to technical knowledge but also include problem-solving abilities, teamwork, communication skills, and understanding of industry practices. This chapter highlights the major skills and knowledge gained during the internship.

4.2 Technical Skills Gained

One of the major outcomes of the internship was the improvement in technical and programming skills. The internship provided hands-on experience in modern web development technologies and tools.

- **Full Stack Development:** Gained practical knowledge of MERN stack technologies including MongoDB, Express.js, React.js, and Node.js. Learned how to build complete web applications by integrating frontend and backend systems.
- **Frontend Development:** Developed user interfaces using HTML, CSS, JavaScript, and React.js. Learned how to create responsive and interactive web pages using component-based architecture.
- **Backend Development:** Learned how to develop server-side logic using Node.js and Express.js. Implemented REST APIs to handle requests and responses between frontend and backend.
- **Database Management:** Gained experience in using MongoDB for storing and managing application data. Learned how to design collections and perform CRUD operations.
- **API Integration:** Understood how to integrate external APIs such as JDoodle for code execution and Stream API for real-time communication.
- **Authentication Systems:** Learned how to implement user authentication using Clerk and JWT-based verification.

- **Debugging and Testing:** Improved debugging skills by identifying and fixing errors in code. Learned how to test applications for performance and functionality.

These technical skills enhanced the ability to develop scalable and efficient web applications.

4.3 Understanding of Software Development Lifecycle

The internship provided a clear understanding of the complete software development lifecycle (SDLC). It helped in learning how real-world software projects are planned, developed, and maintained.

The key stages understood during the internship include:

- **Requirement Analysis** – Understanding project requirements and user needs
- **System Design** – Designing system architecture and workflow
- **Development** – Writing code for frontend and backend
- **Testing** – Identifying and fixing errors in the application
- **Deployment** – Hosting the application on platforms like Vercel and Railway
- **Maintenance** – Updating and improving the application

Understanding the SDLC helped in gaining insight into how professional software development processes are followed in the industry.

4.4 Problem-Solving and Analytical Skills

The internship played a significant role in enhancing problem-solving and analytical thinking abilities. While working on the TalentIQ project, several technical challenges were encountered, such as API errors, debugging issues, and integration problems.

Through continuous practice and guidance from mentors, the ability to analyze problems and find effective solutions improved significantly. Logical thinking skills were enhanced by working on real-time coding tasks and debugging complex issues.

The experience of solving real-world problems helped in developing confidence and the ability to handle challenges independently.

4.5 Professional and Soft Skills

In addition to technical skills, the internship also contributed to the development of important professional and soft skills.

- **Communication Skills:** Improved ability to communicate technical ideas clearly with mentors and team members.
- **Teamwork:** Learned to work collaboratively with other interns and developers on shared tasks.
- **Time Management:** Managed time effectively to complete assigned tasks within deadlines.
- **Adaptability:** Learned to adapt to new technologies and changing requirements.
- **Discipline and Responsibility:** Developed a professional attitude towards work and responsibilities.

These skills are essential for success in any professional environment and play a crucial role in career development.

4.6 Industry Exposure

The internship provided valuable exposure to real-world industry practices and development environments. It helped in understanding how software companies operate and how projects are managed.

Interaction with experienced developers provided insights into industry standards, coding practices, and project workflows. The use of modern tools such as Git, GitHub, and cloud deployment platforms helped in understanding professional development practices.

This exposure helped in bridging the gap between academic knowledge and industry requirements, preparing for future career opportunities in the IT sector.

4.7 Summary of Learning

Overall, the internship at TechieHelp was a highly enriching experience that contributed to both technical and personal growth. It provided a strong foundation in full-stack web development and improved problem-solving abilities.

The knowledge gained during the internship will be beneficial for future academic projects as well as professional career opportunities. The experience has increased confidence and preparedness to work in real-world software development environments.

CHAPTER 5 – CHALLENGES & SOLUTIONS

5.1 Introduction

During the internship at TechieHelp, several challenges were encountered while working on real-world development tasks and the TalentIQ project. These challenges were related to technical implementation, integration of different technologies, debugging issues, and deployment of the application.

Facing these challenges provided valuable learning opportunities and helped in improving problem-solving skills, analytical thinking, and practical knowledge. This chapter describes the major challenges faced during the internship and the solutions implemented to overcome them.

5.2 Technical Challenges

While working on the TalentIQ project, various technical challenges were faced due to the complexity of integrating multiple technologies and handling real-time code execution.

- **Code Execution Issues:** Integrating external APIs such as JDoodle for executing code was challenging. Issues such as incorrect API responses, execution delays, and handling errors were encountered.
- **API Integration Problems:** Connecting the frontend with backend APIs using Axios sometimes resulted in errors such as incorrect endpoints, failed requests, and improper data handling.
- **Authentication and Authorization:** Implementing secure user authentication using Clerk and handling JWT verification required proper understanding of middleware and security practices.
- **Database Connectivity:** Managing MongoDB connections and ensuring proper data storage and retrieval posed challenges, especially during handling multiple user sessions.
- **Real-Time Communication:** Integrating Stream API for chat and real-time communication required handling asynchronous data and managing real-time updates.
- **Debugging Errors:** Identifying and fixing syntax errors, logical bugs, and runtime errors in both frontend and backend code was a frequent challenge.

These challenges required continuous learning and practical implementation to resolve effectively.

5.3 Deployment Challenges

Deployment of the TalentIQ project on cloud platforms also presented several challenges.

- **Environment Configuration:** Setting up environment variables and managing API keys securely during deployment was complex.
- **Frontend and Backend Integration:** Ensuring smooth communication between deployed frontend (Vercel) and backend (Railway) required proper configuration of URLs and APIs.
- **Database Hosting Issues:** Connecting MongoDB Atlas with deployed backend required proper configuration and handling network access settings.
- **Build Errors:** During deployment, build failures occurred due to missing dependencies or incorrect configurations.
- **Performance Issues:** Ensuring efficient execution and minimizing delays during code execution was a key challenge.

These challenges highlighted the importance of understanding deployment processes and cloud-based application hosting.

5.4 Solutions Implemented

To overcome the challenges faced during the internship, several solutions were implemented using proper debugging techniques, research, and guidance from mentors.

- **Resolving Code Execution Issues:** Proper API handling was implemented by validating input data, managing API responses, and handling errors using try-catch blocks.
- **Fixing API Integration Errors:** Axios requests were carefully structured, and correct endpoints were used. Debugging tools and console logs were used to identify issues.
- **Improving Authentication:** Middleware was implemented for secure authentication, and proper token verification methods were used to ensure data security.
- **Database Optimization:** MongoDB queries were optimized, and proper schema design was implemented to manage user sessions efficiently.
- **Handling Real-Time Data:** Asynchronous programming techniques such as async/await were used to manage real-time communication effectively.
- **Debugging Approach:** Systematic debugging methods such as checking logs, analyzing errors, and step-by-step testing were followed to resolve issues.

- **Deployment Fixes:** Environment variables were properly configured, dependencies were installed correctly, and deployment settings were verified to ensure smooth application hosting.

These solutions helped in successfully completing the project and improving technical knowledge.

5.5 Key Learnings from Challenges

The challenges faced during the internship provided valuable learning experiences that contributed significantly to skill development.

- Learned how to handle real-world technical problems
- Improved debugging and troubleshooting skills
- Understood the importance of proper planning and testing
- Gained experience in integrating multiple technologies
- Developed confidence in solving complex problems

These learnings will be beneficial for future projects and professional work in the software development field.

CHAPTER 6 – CONCLUSION & FUTURE SCOPE

6.1 Conclusion

The internship at TechieHelp was a highly valuable and enriching experience that provided practical exposure to real-world software development. It successfully bridged the gap between theoretical knowledge gained during academic studies and its practical implementation in the industry.

During the internship, various modern technologies such as HTML, CSS, JavaScript, React.js, Node.js, Express.js, and MongoDB were learned and applied in real-time projects. The experience of working on the project “TalentIQ – AI-Based Code Execution Platform” helped in understanding full-stack development and the integration of frontend, backend, and database systems.

The internship also provided a clear understanding of the software development lifecycle, including requirement analysis, design, development, testing, and deployment. Working on real-time tasks improved coding skills, debugging abilities, and problem-solving techniques.

In addition to technical knowledge, the internship helped in developing important professional skills such as communication, teamwork, time management, and responsibility. Interaction with mentors and team members enhanced confidence and provided valuable industry insights.

Overall, the internship has significantly contributed to both technical and personal development and has prepared me for future career opportunities in the field of software development.

6.2 Future Scope

The TalentIQ project has significant potential for further development and enhancement. While the current system provides essential functionalities for code execution and session management, several advanced features can be implemented in the future to improve its performance and usability.

Some of the possible future enhancements include:

- Integration of AI-based code suggestions and auto-completion features
- Addition of more programming languages and advanced compiler support
- Implementation of a leaderboard and coding challenge system

- Enhancement of real-time collaboration features for multiple users
- Improved security mechanisms for safe code execution
- Performance optimization for faster execution and response time
- Development of a mobile-friendly or dedicated mobile application

These improvements can transform TalentIQ into a complete coding platform similar to professional coding environments used in the industry.

Furthermore, the knowledge and experience gained during this internship will be highly beneficial for future academic projects, higher studies, and professional career growth in the IT industry.

6.3 Final Remarks

The internship experience at TechieHelp has been a stepping stone towards building a strong career in software development. It provided an opportunity to explore new technologies, work on real-world problems, and gain practical knowledge beyond classroom learning.

The experience has strengthened technical skills, improved confidence, and developed a professional mindset. It has also motivated me to continue learning and exploring new advancements in technology.

This internship will always remain a valuable part of my academic journey and will serve as a foundation for future achievements in the field of computer applications.

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Appendix A – Sample Code

Backend Server Code (server.js)

The server.js file is the main entry point of the backend application. It is developed using Node.js and Express.js and is responsible for initializing the server, configuring middleware, and handling API routes.

```
import express from "express";

import cors from "cors";

import { serve } from "inngest/express";

import { clerkMiddleware } from "@clerk/express";

import { ENV } from "../lib/env.js";

import { connectDB } from "../lib/db.js";

import { inngest, functions } from "../lib/inngest.js";

import chatRoutes from "../routes/chatRoutes.js";

import sessionRoutes from "../routes/sessionRoute.js";

import codeRoutes from "../routes/codeRoutes.js";

const app = express();

// middleware

app.use(express.json());

//  CORS FIX

app.use(

  cors({

    origin: ENV.CLIENT_URL, // your Vercel URL

    credentials: true,

  })

);

// Clerk auth

app.use(clerkMiddleware());

// routes
```

```
app.use("/api/ingest", serve({ client: ingest, functions }));

app.use("/api/chat", chatRoutes);

app.use("/api/sessions", sessionRoutes);

app.use("/api/code", codeRoutes);


// health check

app.get("/health", (req, res) => {

  res.status(200).json({ msg: "api is up and running" });

});

const startServer = async () => {

  try {

    await connectDB();

    app.listen(ENV.PORT, () =>

      console.log("Server is running on port:", ENV.PORT)

    );

  } catch (error) {

    console.error("🔥 Error starting the server", error);

  }

};

startServer();
```

Session Page Implementation (SessionPage.js)

The SessionPage.js file is a core frontend component of the TalentIQ platform, responsible for managing the coding session interface. It is developed using React.js and integrates various functionalities such as problem display, code editor, output panel, and real-time communication.

```
import { useUser } from "@clerk/clerk-react";

import { useEffect, useState } from "react";
```

```
import { useNavigate, useParams } from "react-router";

import { useEndSession, useJoinSession, useSessionById } from "../hooks/useSessions";

import { PROBLEMS } from "../data/problems";

import { executeCode } from "../lib/piston";

import Navbar from "../components/Navbar";

import { Panel, PanelGroup, PanelResizeHandle } from "react-resizable-panels";

import { getDifficultyBadgeClass } from "../lib/utils";

import { Loader2Icon, LogoutIcon, PhoneOffIcon } from "lucide-react";

import CodeEditorPanel from "../components/CodeEditorPanel";

import OutputPanel from "../components/OutputPanel";

import useStreamClient from "../hooks/useStreamClient";

import { StreamCall, StreamVideo } from "@stream-io/video-react-sdk";

import VideoCallUI from "../components/VideoCallUI";

function SessionPage() {

  const navigate = useNavigate();

  const { id } = useParams();

  const { user } = useUser();

  const [output, setOutput] = useState(null);

  const [isRunning, setIsRunning] = useState(false);

  const { data: sessionData, isLoading: loadingSession, refetch } = useSessionById(id);

  const joinSessionMutation = useJoinSession();

  const endSessionMutation = useEndSession();

  const session = sessionData?.session;

  const isHost = session?.host?.clerkId === user?.id;

  const isParticipant = session?.participant?.clerkId === user?.id;

  const { call, channel, chatClient, isInitializingCall, streamClient } = useStreamClient(

    session,

    loadingSession,
```

```
isHost,

isParticipant

);

//  FIX: robust matching (avoids null problemData)

const problemData = session?.problem

? Object.values(PROBLEMS).find(

  (p) => p.title.trim().toLowerCase() === session.problem.trim().toLowerCase()

)

: null;

const [selectedLanguage, setSelectedLanguage] = useState("javascript");

const [code, setCode] = useState(problemData?.starterCode?.[selectedLanguage] || "");

// auto-join session if user is not already a participant and not the host

useEffect(() => {

  if (!session || !user || loadingSession) return;

  if (isHost || isParticipant) return;

  joinSessionMutation.mutate(id, { onSuccess: refetch });

}, [session, user, loadingSession, isHost, isParticipant, id]);

// redirect the "participant" when session ends

useEffect(() => {

  if (!session || loadingSession) return;

  if (session.status === "completed") navigate("/dashboard");

}, [session, loadingSession, navigate]);


// update code when problem loads or changes

useEffect(() => {

  if (problemData?.starterCode?.[selectedLanguage]) {

    setCode(problemData.starterCode[selectedLanguage]);

  }

}
```



```
}, [problemData, selectedLanguage]);

const handleLanguageChange = (e) => {

  const newLang = e.target.value;

  setSelectedLanguage(newLang);

  const starterCode = problemData?.starterCode?.[newLang] || "";

  setCode(starterCode);

  setOutput(null);

};

// 🟢🔥 FINAL FIX (MAIN BUG)

const handleRunCode = async () => {

  if (!problemData?.id) {

    console.error("❌ ProblemId missing", {

      sessionProblem: session?.problem,

      problemData,

    });

    return;

  }

  setIsRunning(true);

  setOutput(null);

  console.log("🚀 FINAL SEND:", {

    language: selectedLanguage,

    problemId: problemData.id,

  });

  const result = await executeCode(

    selectedLanguage,

    code,

    problemData.id // 🟢 FIXED
```

```

);

setOutput(result);

setIsRunning(false);

};

const handleEndSession = () => {

  if (confirm("Are you sure you want to end this session? All participants will be notified.")) {

    endSessionMutation.mutate(id, { onSuccess: () => navigate("/dashboard") });

  }

};

return (

  <div className="h-screen bg-base-100 flex flex-col">

    <Navbar />

    <div className="flex-1">

      <PanelGroup direction="horizontal">

        {/* LEFT PANEL - CODE EDITOR & PROBLEM DETAILS */}

        <Panel defaultSize={50} minSize={30}>

          <PanelGroup direction="vertical">

            {/* PROBLEM DSC PANEL */}

            <Panel defaultSize={50} minSize={20}>

              <div className="h-full overflow-y-auto bg-base-200">

                {/* HEADER SECTION */}

                <div className="p-6 bg-base-100 border-b border-base-300">

                  <div className="flex items-start justify-between mb-3">

                    <div>

                      <h1 className="text-3xl font-bold text-base-content">

                        {session?.problem || "Loading..."}

                      </h1>

                      {problemData?.category && (

```

```

        <p className="text-base-content/60 mt-1">{problemData.category}</p>

    ))

    <p className="text-base-content/60 mt-2">

        Host: {session?.host?.name || "Loading..." } • {" "}

        {session?.participant ? 2 : 1}/2 participants

    </p>

</div>

<div className="flex items-center gap-3">

    <span

        className={`badge badge-lg ${getDifficultyBadgeClass(

            session?.difficulty

        )} `}

    >

        {session?.difficulty

            ? session.difficulty.charAt(0).toUpperCase() +

                session.difficulty.slice(1)

            : "Easy"}

    </span>

    {isHost && session?.status === "active" && (

        <button

            onClick={handleEndSession}

            disabled={endSessionMutation.isPending}

            className="btn btn-error btn-sm gap-2"

        >

            {endSessionMutation.isPending ? (

                <Loader2Icon className="w-4 h-4 animate-spin" />

            ) : (

                <LogoutIcon className="w-4 h-4" />

```

```

    })

    End Session

</button>

})

{session?.status === "completed" && (

    <span className="badge badge-ghost badge-lg">Completed</span>

)}

</div>

</div>

</div>

<div className="p-6 space-y-6">

    {/* problem desc */}

    {problemData?.description && (

        <div className="bg-base-100 rounded-xl shadow-sm p-5 border border-base-300">

            <h2 className="text-xl font-bold mb-4 text-base-content">Description</h2>

            <div className="space-y-3 text-base leading-relaxed">

                <p className="text-base-content/90">{problemData.description.text}</p>

                {problemData.description.notes?.map((note, idx) => (

                    <p key={idx} className="text-base-content/90">

                        {note}

                    </p>

                ))}

            </div>

        </div>

    )}

    {/* examples section */}

    {problemData?.examples && problemData.examples.length > 0 && (

        <div className="bg-base-100 rounded-xl shadow-sm p-5 border border-base-300">

```

```

<h2 className="text-xl font-bold mb-4 text-base-content">Examples</h2>

<div className="space-y-4">

  {problemData.examples.map((example, idx) => (

    <div key={idx}>

      <div className="flex items-center gap-2 mb-2">

        <span className="badge badge-sm">{idx + 1}</span>

        <p className="font-semibold text-base-content">Example {idx + 1}</p>

      </div>

      <div className="bg-base-200 rounded-lg p-4 font-mono text-sm space-y-1.5">

        <div className="flex gap-2">

          <span className="text-primary font-bold min-w-[70px]">

            Input:

          </span>

          <span>{example.input}</span>

        </div>

        <div className="flex gap-2">

          <span className="text-secondary font-bold min-w-[70px]">

            Output:

          </span>

          <span>{example.output}</span>

        </div>

        {example.explanation && (

          <div className="pt-2 border-t border-base-300 mt-2">

            <span className="text-base-content/60 font-sans text-xs">

              <span className="font-semibold">Explanation:</span>{" "}

              {example.explanation}

            </span>

          </div>

        )}

      </div>

    )}

  )}

```

```

        })

        </div>

        </div>

        )))

        </div>

        </div>

    })

    { /* Constraints */

    {problemData?.constraints && problemData.constraints.length > 0 && (

    <div className="bg-base-100 rounded-xl shadow-sm p-5 border border-base-300">

    <h2 className="text-xl font-bold mb-4 text-base-content">Constraints</h2>

    <ul className="space-y-2 text-base-content/90">

    {problemData.constraints.map((constraint, idx) => (

    <li key={idx} className="flex gap-2">

    <span className="text-primary">•</span>

    <code className="text-sm">{constraint}</code>

    </li>

    )))

    </ul>

    </div>

    })

    </div>

    </div>

    </Panel>

    <PanelResizeHandle className="h-2 bg-base-300 hover:bg-primary transition-colors cursor-row-resize" />

    <Panel defaultSize={50} minSize={20}>

```

```

<PanelGroup direction="vertical">

  <Panel defaultSize={70} minSize={30}>

    <CodeEditorPanel

      selectedLanguage={selectedLanguage}

      code={code}

      isRunning={isRunning}

      onLanguageChange={handleLanguageChange}

      onCodeChange={(value) => setCode(value)}

      onRunCode={handleRunCode}

      currentProblem={problemData} //  important

    />

  </Panel>

  <PanelResizeHandle className="h-2 bg-base-300 hover:bg-primary transition-colors cursor-
row-resize" />

  <Panel defaultSize={30} minSize={15}>

    <OutputPanel output={output} />

  </Panel>

</PanelGroup>

</Panel>

</PanelGroup>

</Panel>

  <PanelResizeHandle className="w-2 bg-base-300 hover:bg-primary transition-colors cursor-col-
resize" />

  { /* RIGHT PANEL */ }

  <Panel defaultSize={50} minSize={30}>

    <div className="h-full bg-base-200 p-4 overflow-auto">

      {isInitializingCall ? (

        <div className="h-full flex items-center justify-center">

```

```

        <Loader2Icon className="w-12 h-12 mx-auto animate-spin text-primary mb-4" />

        <p className="text-lg">Connecting to video call...</p>

    </div>

    ): !streamClient || !call ? (

        <div className="h-full flex items-center justify-center">

            <PhoneOffIcon className="w-12 h-12 text-error" />

        </div>

    ): (

        <StreamVideo client={streamClient}>

            <StreamCall call={call}>

                <VideoCallUI chatClient={chatClient} channel={channel} />

            </StreamCall>

        </StreamVideo>

    )}

</div>

</Panel>

</PanelGroup>

</div>

</div>

);

}

export default SessionPage;

```


Appendix B – Internship Certificate

The internship completion certificate provided by the organization is attached in this section as proof of successful completion of the internship training.



Figure 14: Internship Certificate